23k-6005
BSAI-7A
Hania Ahmed

AGENTIC AI ASSIGNMENT BETA Date: _____

Q3,

$z_t = [5, 3, 2, 0]$

$z_s = [3, 2.5, 1, 0.5]$

$y = [1, 0, 0, 0]$

$T = 2, \alpha = 0.5, \eta = 0.5$

(a) Teacher and student Probabilities at $T=1$

$$P_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

For the teacher:

$$\sum_j e^{z_t} = e^5 + e^3 + e^2 + e^0$$

$$= 148.4132 + 20.0855 + 7.3891 + 1$$

$$= 176.8878$$

Therefore, $p_1 = [0.83902, 0.11355, 0.04177, 0.00565]$

for the student:

$$\sum_j e^{z_s} = e^3 + e^{2.5} + e^1 + e^{0.5}$$

$$= 20.0855 + 12.1825 + 2.7183 + 1.6487$$

$$= 36.6350.$$

Therefore, $q_1 = [0.54826, 0.33254, 0.07420, 0.04500]$

(b) Softened Probabilities at $T=2$

$$P_i(T) = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}$$

Teachers scaled logits:

$z_t/2 = [2.5, 1.5, 1, 0]$

$p_2 = [0.59769, 0.21988, 0.13336, 0.04906]$

Students scaled logits:

$z_s/2 = [1.5, 1.25, 0.5, 0.25]$

$q_2 = [0.41098, 0.32007, 0.15119, 0.11775]$

(c) Calculate KL Divergence

$$KL(p_2 \| q_2) = \sum_i p_{2,i} \ln \frac{p_{2,i}}{q_{2,i}}$$

$$KL = 0.59769 \ln \frac{0.59769}{0.41098} + 0.21988 \ln \frac{0.21988}{0.32007} + 0.13336 \ln \frac{0.13336}{0.15119} + 0.04906 \ln \frac{0.04906}{0.11775}$$

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KL = 0.081607

(d) Hard label cross Entropy - only Hospital correct word, first student probability matters.

$$CE = -\sum_i y_i \ln q_{s,i}$$

$$CE = -\ln(0.54826) = 0.60100$$

(e) Total Distillation loss

$$L = \alpha T^2 KL + (1-\alpha) CE$$

$$L = 0.5(2)^2 (0.081607) + 0.5(0.601005)$$

$$L = 0.163215 + 0.300502$$

$$L = 0.463717$$

(f) Gradient w.r.t students logit

$$\frac{\partial L}{\partial z_s} = \alpha T (q_s - p_t) + (1-\alpha)(q_s - y)$$

Substituting $T=2$ $\alpha = 0.5$:

$$\nabla L = (q_s - p_t) + 0.5(q_s - y)$$

For each logit:

$g_1 = [0.41098 - 0.59769] + 0.5(0.54826 - 1)$ $g_1 = -0.41258$

$g_2 = (0.32007 - 0.21988) + 0.5(0.33254)$ $g_2 = 0.26646$

$g_3 = (0.15119 - 0.13336) + 0.5(0.07420)$ $g_3 = 0.05493$

$g_4 = (0.11775 - 0.04906) + 0.5(0.07500)$ $g_4 = 0.09119$

$\nabla L = [-0.41258, 0.26646, 0.05493, 0.09119]$

(g) Updated logits and new loss

Gradient descent: $z_s^{new} = z_s - \eta \nabla L$

using $\eta = 0.5$:

$z_1^{new} = 3 - 0.5(-0.41258) = 3.20629$

$z_2^{new} = 2.5 - 0.5(0.26646) = 2.36677$

$z_3^{new} = 1 - 0.5(0.05493) = 0.97254$

$z_4^{new} = 0.5 - 0.5(0.09119) = 0.45441$

$z_s^{new} = [3.20629, 2.36677, 0.97254, 0.45441]$

$T = 1 = e^{3.2062} + e^{2.3668} + e^{0.9726} + e^{0.4544}$ $\$\$ = 39.5708$

$$q_{s,new} = \left[\frac{24.6876}{39.5708}, \frac{10.6632}{39.5708}, \frac{2.6448}{39.5708}, \frac{1.5752}{39.5708}\right]$$

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$q_{i,new}$ = [0.6239, 0.2695, 0.06684, 0.039981]

T=2, z_i = [1.6032, 1.1834, 0.4863, 0.2772]

$e^{1.6032} + e^{1.1834} + e^{0.4863} + e^{0.2772} = 11.1157$

$q_{i,new} = \left[\frac{4.9689}{11.1157}, \frac{3.2659}{11.1157}, \frac{1.6263}{11.1157}, \frac{1.2551}{11.1157}\right]$

= [0.447, 0.2938, 0.1463, 0.1129]

KL:

"Hospital" = 0.5977 ln (0.5977 / 0.447) = 0.1736.

"Clinic" = 0.2199 ln (0.2199 / 0.2938) = -0.06371

"Ward" = 0.1337 ln (0.1337 / 0.1463) = -0.01231

"School" = 0.04906 ln (0.04906 / 0.1129) = -0.04089

KL = 0.1736 - 0.06371 - 0.01231 - 0.04089

KL = 0.05669

CE = -ln(0.6239), 0.47718

Loss = (0.5)(2)^2(0.05669) + 0.5 (0.47718)

Loss = 0.3793.

Q4, Stage1 Accuracy and 95% confidence interval - Part 1

Number of correct predictions.

100 + 80 + 50 = 230

Total reviews = 300

Accuracy = $\frac{230}{300}$ = 0.7667 Accuracy = 76.67 %

Wilson score method z = 1.96, n = 300, $\hat{p}$ = 0.7667, for 95% confidence interval

Wilson centre:

$$C = \frac{\hat{p} + \frac{z^2}{2n}}{1 + \frac{z^2}{n}}$$

$$C = \frac{0.7667 + \frac{1.96^2}{600}}{1 + \frac{1.96^2}{300}} = 0.7633$$

Wilson half width:

$$h = \frac{1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{1.96^2}{4n^2}}}{1 + \frac{1.96^2}{n}} \quad h = 0.04768$$

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Therefore 95% CI = [0.8156, 0.8113]

Precision, Recall, and F1 for Stage 2 - Part 2

Precision = $\frac{TP}{TP+FP}$, Recall = $\frac{TP}{TP+FN}$, F1 = $\frac{2PR}{P+R}$

Positive class: TP = 100, FP = 8+20 = 28, FN = 5+15 = 20

P = $\frac{100}{128}$ = 0.78125, R = $\frac{100}{120}$ = 0.83333, F1 = 0.80645

Negative class: TP = 80, FP = 5+20+15 = 40, FN = 8+12 = 20

P = $\frac{80}{95}$ = 0.84211, R = $\frac{80}{100}$ = 0.80000, F1 = 0.82051

Neutral class: TP = 50, FP = 15+12 = 27, FN = 20+10 = 30

P = $\frac{50}{77}$ = 0.64935, R = $\frac{50}{80}$ = 0.62500, F1 = 0.63694

Macro, weighted and micro F1 - Part 3

Macro F1: $F_{macro} = \frac{0.80645 + 0.82051 + 0.63694}{3}$, $F_{macro} = 0.75464$

Weighted F1: $F_{weighted} = \frac{120(0.80645) + 100(0.82051) + 80(0.63694)}{300}$, $F_{weighted} = 0.756$

micro F1: For single label multi class classification, micro F1 equals overall accuracy.

$F_{micro} = 230/300$, $F_{micro} = 0.76667$

Interpreting Stage 2 - Part 4

The zero shot model performs better on Positive and Negative reviews than on Neutral reviews. It correctly classifies only 50 out of 80 neutral reviews. Some mixed reviews are incorrectly classified as Positive or Negative. Giving the model a clear definition of Neutral and a few examples of mixed reviews would help improve its predictions.

STAGE 3 EVALUATION - PART 5

Stage 2 has: 106 + 84 + 64 = 254

correct predictions: Accuracy = 254/300 = 0.8467 = Accuracy = 84.67%.

Using same wilson method.

centre: $\frac{0.8467 + (1.96^2/600)}{1 + (1.96^2/300)} = 0.8413$

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Half width: $1.96 \sqrt{0.8467(1-0.8467)/300 + (1.96^2/4{,}300)} \Big/ 1 + \left(\frac{1.96^2}{300}\right)$

$= 1.96 \times 0.0211 / 1.0128 = 0.04083$

95% wilson CI: 0.8433 ± 0.04083

$= [0.80153, 0.88303]$

Positive class: TP = 106, FP = 6+9 = 15, FN = 4+10 = 14

$P = \frac{106}{121} = 0.87603$, $R = \frac{106}{120} = 0.8833$, $F1 = 0.87967$

Negative class: TP = 84, FP = 4+7 = 11, FN = 6+10 = 16

$P = \frac{84}{95} = 0.88421$, $R = \frac{84}{100} = 0.84$, $F1 = 0.86154$

Neutral: TP = 64, FP = 10+10 = 20, FN = 9+7 = 16.

$P = \frac{64}{84} = 0.76190$, $R = \frac{64}{80} = 0.80$, $F1 = 0.78049$

Calculating Averages:

$F_{macro} = \frac{0.87967 + 0.86154 + 0.78049}{3} \Rightarrow F_{macro} = 0.84056$

$F_{weighted} = \frac{120(0.87967) + 100(0.86154) + 80(0.78049)}{300} \Rightarrow F_{weighted} = 0.84718$

$F_{micro} = 254/300$ $F_{micro} = 0.84667$

metric	Stage 1	Stage 2
Accuracy	0.7667	0.8467
macro F1	0.7546	0.8406
weighted F1	0.7669	0.8472
Micro F1	0.7667	0.8467
Neutral Recall	0.6250	0.8000

The improved prompt increases accuracy from 76.67% to 84.67%. The largest improvement is in the neutral class, where recall increases from 67.5% to 80%. This shows that defining Neutral clearly and adding adding examples helps the model recognize mixed reviews.

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Part 6 - Prompt Stability

Given $x = [0.847, 0.840, 0.853, 0.827, 0.850]$

mean: $\bar{x} = \frac{0.847 + 0.840 + 0.853 + 0.827 + 0.850}{5}$

$\bar{x} = 0.8457$

Sample Standard Deviation:

$$s = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{n-1}}$$

$s = \sqrt{\frac{0.0001812}{4}}$ $s = 0.00673$

Report: 0.8457 ± 0.0067

The accuracy changes slightly when the examples are placed in a different order. However, the standard deviation is small, which suggests that the prompt is fairly stable across these five runs.

Part 7 - BLEU-4, ROUGE, and METEOR

Reference: the delivery was late but the product quality is good

Candidate: the product quality is good but delivery was slow.

Reference length: 10 words

Candidate length: 9 words

BLEU-4:

$p_1 = \frac{8}{9}$, $p_2 = \frac{5}{8}$, $p_3 = \frac{3}{7}$, $p_4 = \frac{2}{6}$

Brevity Penalty: $BP = e^{1 - 10/9} = 0.89484$

BLEU-4: $BLEU_4 = BP(p_1 p_2 p_3 p_4)^{1/4}$

$= 0.89484 (8/9 \times 5/8 \times 3/7 \times 2/6)^{1/4}$

$BLEU_4 = 0.47495$

ROUGE-1:

$P = \frac{8}{9} = 0.88889$, $R = \frac{8}{10} = 0.8$ $F_1 = \frac{2(0.88889)(0.8)}{0.88889 + 0.8} = 0.84211$

ROUGE-2:

$P = \frac{5}{8} = 0.625$, $R = \frac{5}{9} = 0.55556$ ROUGE-2: 0.5882

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ROUGE-2

The largest common subsequence is:

The product quality is good

so LCS length = 5.

$P = \frac{5}{9}$, $R = \frac{5}{10}$ ROUGE-L : 0.52632

METEOR : using exact word matches

matching words = 8, Candidate length = 9, Reference length = 9, matching chunks = 3

$P = \frac{8}{9}$, $R = \frac{8}{10}$

weighted Harmonic mean : $F = \frac{10PR}{R+9P}$ $F = 0.80808$

Recent- order.

city stable

fragmentation Penalty: $Penalty = 0.5\left(\frac{3}{8}\right)^3$: Penalty : 0.02637

METEOR : METEOR = F (1 - Penalty)

: 0.80808 (1 - 0.02637)

METEOR = 0.78677

PART 8 - PERPLEXITY :

Given Probabilities: [0.60, 0.40, 0.50, 0.80, 0.70, 0.30, 0.25, 0.50, 0.20]

Perplexity $PPL = \exp\left(-\frac{1}{N}\sum \ln p_i\right)$

For 8 tokens:

$\sum \ln p_i = \ln(0.60) + \ln(0.10) + \ln(0.50) + \ln(0.80) + \ln(0.70) + \ln(0.30) + \ln(0.25$

$+ \ln(0.50) + \ln(0.20)$

$\sum \ln p_i = -7.59293$

$PPL = e^{7.59293/9} = 2.32486$

The perplexity is approximately 2.32. This indicates how confidently the model predicts its generated tokens. A lower perplexity usually means greater confidence, but it does not guarantee that the generated explanation is factually correct.

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